

LAB EXPERIMENT RECORD

SET 8 — Employee Performance Analysis

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	3 October 2026
Day	Saturday

A. Aim

To analyze employee performance using line charts, bar charts, tables, and an interactive Tableau dashboard.

B. Dataset

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Program 1 — Line Chart of Employee Performance

```
employee_id <- c(1, 2, 3)
```

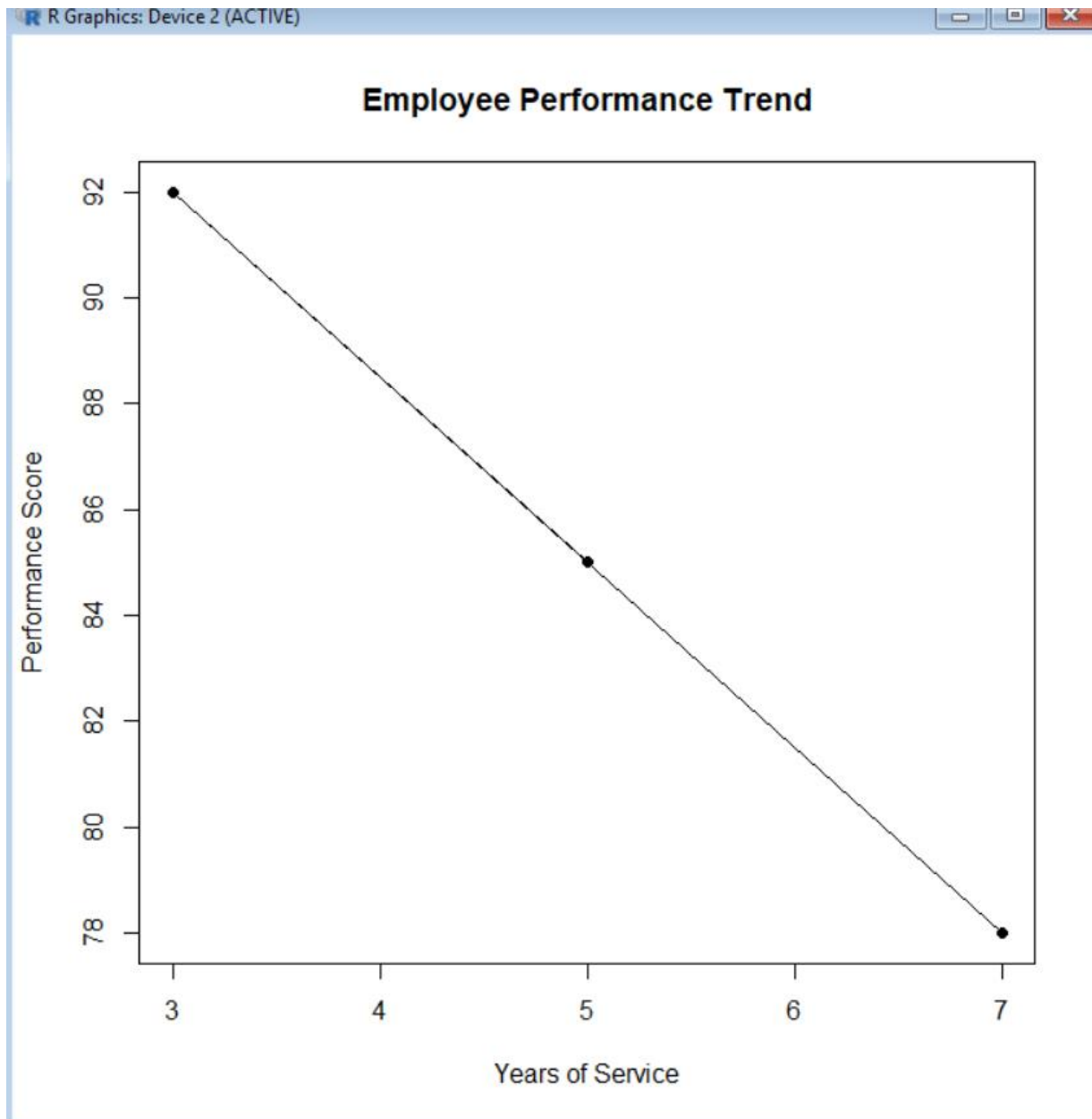
```
years <- c(5, 3, 7)
```

```
performance <- c(85, 92, 78)
```

```
employee <- data.frame(  
  employee_id,  
  years,  
  performance  
)
```

```
plot(  
  employee$years,  
  employee$performance,  
  type = "o",  
  main = "Employee Performance Trend",  
  xlab = "Years of Service",  
  ylab = "Performance Score",  
  pch = 19  
)
```

```
text(  
  employee$years,  
  employee$performance,  
  labels = employee$employee_id,  
  pos = 3  
)
```



Observation

The employee with **3 years of service** has the **highest performance score of 92**, while the employee with 7 years has a score of 78.

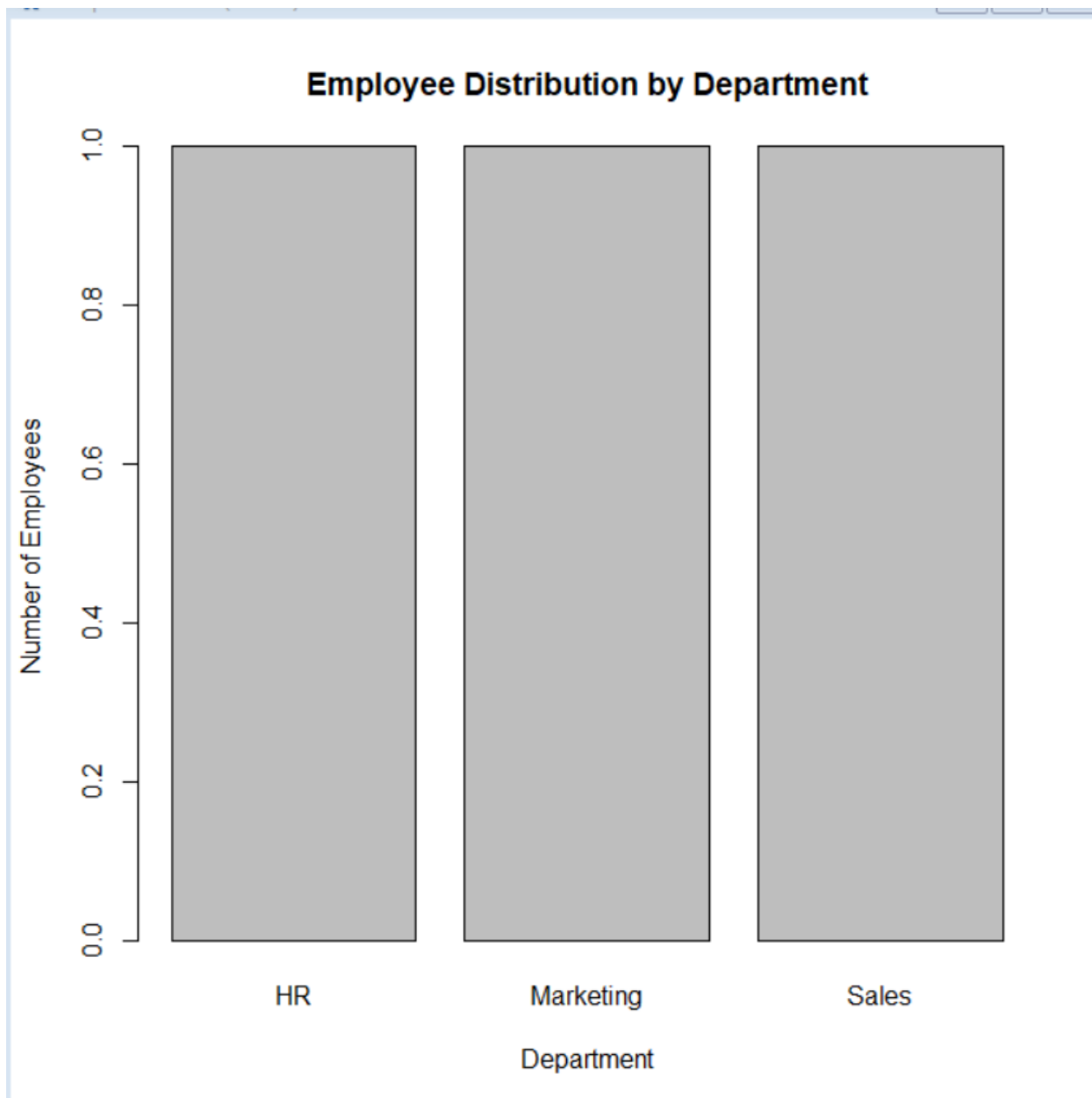
Program 2 — Bar Chart of Employees by Department

```
department <- c("Sales", "HR", "Marketing")
```

```
department_count <- table(department)
```

```
barplot(  
  department_count,  
  main = "Employee Distribution by Department",  
  xlab = "Department",  
  ylab = "Number of Employees"  
)
```

```
text(  
  seq_along(department_count),  
  department_count,  
  labels = department_count,  
  pos = 3  
)
```



Observation

Each department has **one employee** in the given dataset.

Program 3 — Employee Performance Table

```
employee_data <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)  
  
print(employee_data)  
  
print(employee_data)  
Employee_ID Department Years_of_Service Performance_Score  
      1      Sales              5             85  
      2        HR              3             92  
      3  Marketing              7             78  
|
```

Observation

The table displays the employee ID, department, years of service, and performance score for each employee.

Program 4 — Tableau Dashboard

Line Chart

1. Import the dataset into Tableau.
2. Drag **Years of Service** → Columns.
3. Drag **Performance Score** → Rows.
4. Select **Line Chart**.
5. Add title: **Employee Performance Trend**.

Bar Chart

1. Create a new worksheet.
2. Drag **Department** → Columns.
3. Drag **Employee ID** → Rows.
4. Select **Bar Chart**.
5. Turn on **Show Mark Labels**.

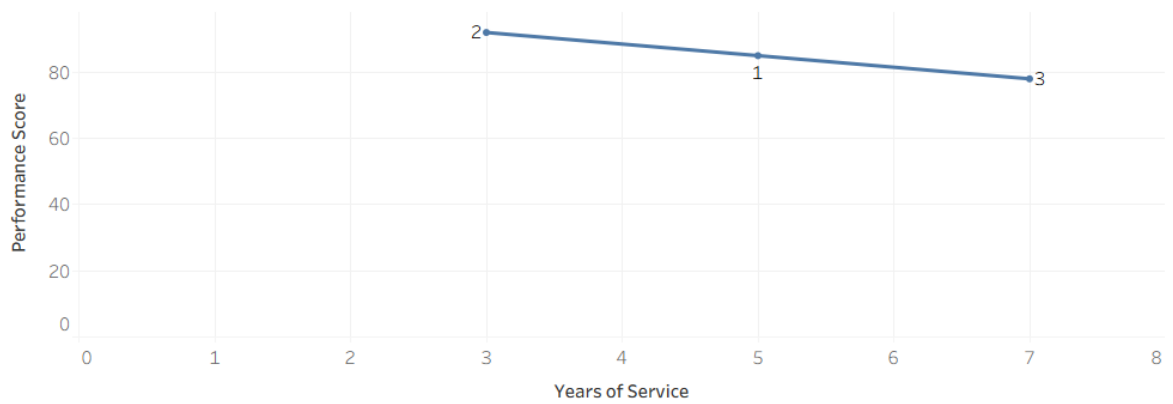
Table

1. Create another worksheet.
2. Drag **Employee ID, Department, Years of Service, Performance Score** into the view.
3. Select **Text Table**.

Dashboard

1. Select **New Dashboard**.
2. Add the line chart.
3. Add the department bar chart.
4. Add the employee table.
5. Add filters such as Department if required.

Employee Performance Trend



Employee Performance Table

Department	Employee	Years of Service	Performance Score
HR	2	5	50
		5	50
Marketing	3	5	50
		5	50
Sales	1	5	50
		5	50

Employee Distribution by Department

Employee	HR	Marketing	Sales
1			
2			
3			

E. Overall Procedure

1. Enter employee performance data.
2. Plot employee performance against years of service.
3. Create a department-wise employee bar chart.
4. Display employee information in a table.
5. Combine all visualizations in Tableau.
6. Add filters for interactive exploration.

F. Result

Employee performance data was successfully visualized using **line charts, bar charts, tables, and a Tableau dashboard.**

G. Key R Functions Used

Function	Purpose
c()	Create vectors
data.frame()	Create dataset
plot()	Create line chart
barplot()	Create bar chart
table()	Count departments
text()	Add labels
print()	Display table